

Coding & Solution

DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor
MAXIMUM MARKS	5 MARKS

1. Overview

This document describes the implementation logic, preprocessing steps, and algorithm selection for the Human Development Index (HDI) Prediction System.

2. Core Preprocessing Logic

Before training the model, the data pipeline cleans the features in the dataset to ensure stability.

2.1 Feature Selection

The dataset `data/hdi\_data.csv` contains multiple columns. The preprocessing pipeline selects the five most relevant features:

1. `Country` (Categorical)
2. `Life\_Expectancy` (Continuous)
3. `Expected\_Years\_Schooling` (Continuous)
4. `Mean\_Years\_Schooling` (Continuous)
5. `GNI\_Per\_Capita` (Continuous)

2.2 Missing Value Imputation

If any indicators are missing, the pipeline fills them using mean imputation:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

This fills null entries with the overall feature average, preventing training errors.

2.3 Categorical Encoding

The `Country` column contains names that must be converted to numeric format. We fit a `LabelEncoder` to convert these country names into integer indices (0 to \$K-1\$). This encoder is saved as `models/encoder.pkl` to ensure the same mapping is used during prediction.

3. Algorithm Selection & Logic

3.1 Linear Regression

We use a Linear Regression model to predict the continuous HDI score:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

Where:

- $\hat{y}$ is the predicted HDI score.
- x_i are the input features (Life Expectancy, EYS, MYS, GNI, Encoded Country).
- θ_i are the learned parameters (coefficients).
- θ_0 is the intercept.

3.2 Evaluation Metrics

- **Root Mean Squared Error (RMSE):** Measures the average magnitude of the prediction error:

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_{j=1}^M (y_j - \hat{y}_j)^2}$$

- **Coefficient of Determination (R^2):** Measures the proportion of variance explained by the model:

$$R^2 = 1 - \frac{\sum (y_j - \hat{y}_j)^2}{\sum (y_j - \bar{y})^2}$$

The model achieves an $R^2 = 0.8834$ and an $\text{RMSE} = 0.0361$, indicating highly accurate predictions.

4. Web Implementation & Request Lifecycle

The Flask application processes requests as follows:

```
[User Browser] (Enters parameters in indexnew.html)
|
▼ (HTTP POST /predict with form fields)
[Flask Server] (app.py)
|
├─ 1. Cast parameters to float/int
└─ 2. Load encoder.pkl and transform country label
```

```
|→ 3. Load hdi_model.pkl and run model.predict()
|→ 4. Clamp predicted score to [0.0, 1.0]
|→ 5. Classify score into tier (classify_hdi)
|→ 6. Fetch tier recommendations (get_hdi_insights)
|
▼ (HTML with results, badge classes, and gauge parameters)
[User Browser] (Renders result card, gauge fill, and recommendations)
```